AJAY.I - 71812401007 · FOUNDATIONS OF AGENTIC AI SYSTEMS · MODULE 3 — MEMORY SYSTEMS AND EMERGING TRENDS NOTES

1. MEMORY SYSTEMS FOR AI AGENTS

1.1 What is Memory in AI Agents?

Simple explanation:

Memory is nothing but the ability of an AI agent to remember information and use it when needed.

For example, imagine you are asking an AI agent to help you study.

- Today, you tell it that you are studying Agentic AI.
- Tomorrow, you ask it to continue your preparation.
- If it remembers your previous topics, it can continue without asking everything again.

This is the basic purpose of memory in AI agents.

Technical explanation:

Memory is a mechanism that allows an intelligent agent to store, retrieve, update, and use information from previous interactions, observations, and experiences to improve future decision-making.

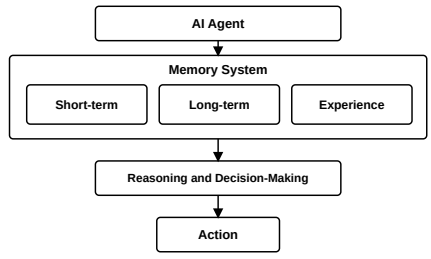
Memory helps agents maintain context, learn from past experiences, and perform tasks more effectively.

Why does an AI agent need memory?

1. To remember previous conversations.
2. To maintain context during long tasks.
3. To store useful information about the environment.
4. To learn from previous experiences.
5. To avoid repeating the same mistakes.
6. To make better decisions using past information.

Basic architecture of agent memory

DIAGRAM



Basic architecture of agent memory

The agent retrieves relevant information from memory and uses it to make decisions and perform actions.

2. SHORT-TERM AND LONG-TERM MEMORY

Memory in AI agents is commonly classified into short-term memory and long-term memory.

2.1 Short-term Memory

Simple explanation:

Short-term memory is nothing but remembering information that is needed right now.

For example, when you ask an AI agent:

"Calculate 25 × 4 and then add 10."

The agent needs to remember the result of the first calculation while performing the second calculation.

It does not need to remember this information forever.

Technical explanation:

Short-term memory stores temporary information required for the current task, conversation, or reasoning process.

It is usually limited in capacity and may be cleared or replaced when the task or conversation ends.

Characteristics

- Stores temporary information.
- Maintains the current conversation context.
- Helps perform multi-step tasks.
- Has limited capacity.
- Frequently updated.
- Information may disappear after the session.

Example

CURRENT CONVERSATION

User: My name is Ajay.

User: I am studying Agentic AI.

User: Explain memory systems.

Short-term memory stores:

- Name: Ajay
- Current subject: Agentic AI
- Current topic: Memory systems

The agent uses this information to answer the current question correctly.

2.2 Long-term Memory

Simple explanation:

Long-term memory is nothing but remembering information for a long period.

For example, an AI agent may remember that a user prefers Python programming instead of Java.

Even after several conversations, the agent can use this information.

Technical explanation:

Long-term memory stores important information, knowledge, preferences, and experiences that may be useful across multiple tasks or sessions.

It is generally stored in external databases, files, vector databases, or persistent memory systems.

Characteristics

- Stores information for a long duration.
- Can retain information across sessions.
- Supports personalized responses.
- Stores important facts and experiences.
- Can retrieve information when required.
- Usually requires a storage and retrieval mechanism.

Example

An AI learning assistant stores:

```
User prefers Python.
User is studying Computer Science.
User wants simple explanations.
```

When the user asks for programming help later, the agent can use these stored preferences.

2.3 Difference between Short-term and Long-term Memory

Short-term memory	Long-term memory
Stores temporary information.	Stores information for long periods.
Used in the current task.	Used across different tasks.
Limited capacity.	Can use external storage for large amounts of data.
Frequently updated.	Updated when important information changes.
Example: Current conversation.	**Example:** User preferences and past experiences.

Important exam point: Short-term memory maintains the current context, whereas long-term memory preserves useful knowledge and experiences for future tasks.

3. RETRIEVAL-AUGMENTED GENERATION (RAG)

3.1 What is RAG?

Simple explanation:

RAG is nothing but giving an AI agent the ability to search for information before answering a question.

Imagine you are writing an exam.

You do not remember the complete answer, so you open your textbook, find the relevant topic, read it, and then write the answer.

RAG works similarly.

Instead of answering only from its existing knowledge, the AI searches an external document or database and uses the retrieved information to generate an answer.

Technical explanation:

Retrieval-Augmented Generation (RAG) is an AI technique that combines information retrieval with language generation.

It retrieves relevant information from an external knowledge source and provides that information to a language model as context for generating a response.

RAG improves factual accuracy, allows access to updated information, and reduces dependence on information stored inside the model during training.

3.2 Why is RAG required?

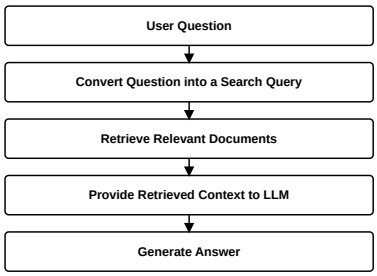
A normal language model may have the following limitations:

1. It may not know recently updated information.
2. It may generate incorrect answers.
3. It may not have access to private documents.
4. It may not remember large amounts of external information.
5. It may produce answers without supporting evidence.

RAG addresses these problems by retrieving relevant information from external sources.

3.3 Architecture of RAG

DIAGRAM



Architecture of RAG

3.4 Working of RAG

RAG works through the following steps.

Step 1: User Query

The user asks a question.

Example:

"What are the memory systems used in Agentic AI?"

Step 2: Query Processing

The system processes the question and identifies the information required.

Step 3: Information Retrieval

The retrieval system searches relevant documents, PDFs, databases, or other knowledge sources.

Step 4: Context Selection

The system selects the most relevant information and provides it to the language model.

Step 5: Answer Generation

The language model uses the retrieved information and its reasoning capabilities to generate an answer.

Step 6: Response

The generated answer is returned to the user.

3.5 Important components of RAG

Component	Purpose
Knowledge source	Stores documents and information.
Document chunking	Divides large documents into smaller pieces.
Embedding model	Converts text into numerical vectors.
Vector database	Stores and searches vector representations.
Retriever	Finds relevant information.
Language model	Generates the final answer.

What is an embedding?

Simple explanation: An embedding is nothing but converting text into numbers so that a computer can understand its meaning and compare it with other text.

For example:

- "AI agent memory"
- "Memory used by intelligent agents"

These sentences have similar meanings.

An embedding model converts them into numerical vectors that may be close to each other in vector space.

This helps the retrieval system find related information even when the exact words are different.

Applications of RAG

- Question answering systems.
- Document-based chatbots.
- University learning assistants.
- Healthcare information systems.
- Enterprise knowledge search.
- Customer support systems.

Example: A university chatbot uses RAG to search college regulations and answer student questions based on official documents.

4. MEMGPT AND OS-LIKE MEMORY MANAGEMENT

4.1 What is MemGPT?

Simple explanation:

MemGPT is nothing but a system that helps an AI manage its memory like a computer operating system manages its memory.

Imagine your computer has limited RAM.

You cannot keep every application and every file inside RAM.

Therefore, the operating system moves information between RAM and storage whenever necessary.

MemGPT applies a similar idea to AI agents.

The agent keeps important information in its active context and moves other information into external memory.

Technical explanation:

MemGPT is a memory-management architecture for language models that uses an operating-system-inspired approach to manage limited context capacity.

It allows an AI agent to maintain information beyond its immediate context by managing data between active context and external storage.

It supports long conversations, memory retrieval, and context management.

4.2 Why is MemGPT needed?

Language models have a limited context window.

The context window is the amount of information the model can process at one time.

When a conversation becomes very long, the model may not be able to keep every previous message in its active context.

MemGPT helps by storing information externally and retrieving it when required.

4.3 OS-like memory management

A computer operating system uses:

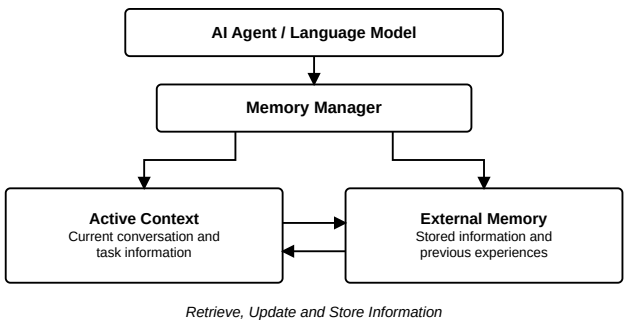
- RAM for currently required information.
- Disk storage for information that is not currently active.
- Memory management to move information between storage locations.

MemGPT uses a similar architecture.

Operating system	MemGPT
RAM	Active context
Disk storage	External memory
Memory manager	Memory-management logic
Load data into RAM	Retrieve information into context
Save data to disk	Store information externally

4.4 MemGPT architecture

DIAGRAM



MemGPT architecture

4.5 Working of MemGPT

1. The agent receives a user request.

2. It checks the information available in its active context.
3. If required information is missing, it searches external memory.
4. Relevant information is loaded into the active context.
5. The agent performs reasoning using the retrieved information.
6. Important new information is stored in external memory.
7. The agent continues the task.

Example

A user has a long conversation with an AI agent about a software project.

The agent cannot keep every message inside its active context.

MemGPT can store previous project details externally and retrieve relevant information when the user asks about earlier decisions.

Advantages

- Supports long conversations.
- Reduces context limitations.
- Helps maintain important information.
- Enables memory retrieval.
- Supports persistent agent workflows.

5. AGENTIC AI CAPABILITIES

5.1 What is Agentic AI?

Simple explanation:

Agentic AI is nothing but AI that can work toward a goal instead of only answering a question.

For example:

A normal AI answers:

"How can I book a train ticket?"

An agentic AI can:

1. Understand the travel requirement.
2. Search available trains.
3. Compare options.
4. Select a suitable train.
5. Ask for required confirmation.
6. Perform the permitted booking action.

The important difference is that an agentic system can perform multiple steps to achieve a goal.

Technical explanation:

Agentic AI refers to AI systems capable of reasoning, planning, using tools, interacting with environments, and executing actions to achieve specified objectives with varying levels of autonomy.

5.2 Major capabilities of Agentic AI

1. Perception

Simple explanation: Understanding what is happening around the agent.

Technical explanation: Perception allows an agent to collect and interpret information from its environment using text, images, audio, sensors, or other inputs.

Example: A robot camera detects an obstacle.

2. Reasoning

Simple explanation: Thinking about what should be done.

Technical explanation: Reasoning enables an agent to analyze information, understand relationships, evaluate alternatives, and make decisions.

Example: The agent decides which route is shortest.

3. Planning

Simple explanation: Creating a sequence of steps to reach a goal.

Technical explanation: Planning determines the actions and their order required to achieve an objective.

Example: Plan a journey from Coimbatore to Chennai.

4. Memory

Simple explanation: Remembering useful information.

Technical explanation: Memory allows the agent to store and retrieve previous observations, knowledge, and experiences.

Example: Remembering the user's travel preferences.

5. Tool Use

Simple explanation: Using external applications to perform work.

Technical explanation: Tool use enables an agent to interact with APIs, databases, browsers, calculators, and other software systems.

Example: Using a weather API to retrieve current weather.

6. Learning

Simple explanation: Improving based on previous experiences.

Technical explanation: Learning allows an agent to update its knowledge or behavior based on feedback, observations, or training.

Example: A robot improves its navigation through experience.

7. Autonomous Action

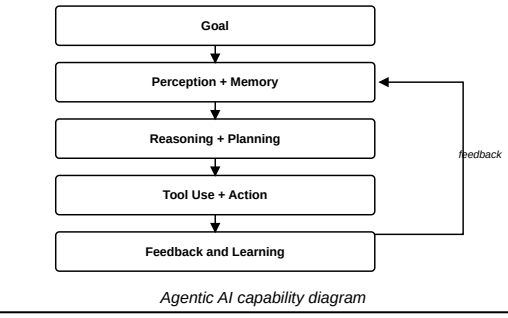
Simple explanation: Performing actions without requiring the user to provide every individual instruction.

Technical explanation: Autonomy enables an agent to execute planned actions independently within its permitted scope.

Example: An agent automatically checks a system and reports detected failures.

5.3 Agentic AI capability diagram

DIAGRAM



Agentic AI capability diagram

The agent observes the environment, decides what to do, performs actions, and uses feedback to improve its next decision.

Applications

- Autonomous robots.
- Personal AI assistants.
- Software development agents.
- Smart manufacturing.
- Healthcare support systems.
- Autonomous vehicles.
- Customer service automation.

6. GENERAL PURPOSE AGENTS (GPA)

6.1 What is a General Purpose Agent?

Simple explanation:

A General Purpose Agent is nothing but an AI agent that can perform different types of tasks instead of being designed for only one specific task.

For example, the same agent may:

- Write Python code.
- Search for information.
- Analyze documents.
- Create reports.
- Use external tools.

It can handle different tasks depending on the user's requirements.

Technical explanation:

A General Purpose Agent (GPA) is an AI system designed to perform a broad range of tasks across different domains by combining reasoning, planning, memory, tool use, and interaction with environments.

Unlike task-specific agents, GPAs are not restricted to a single predefined application.

6.2 Types of AI agents

The traditional classification of AI agents includes the following:

Type	Explanation
Simple Reflex Agent	Selects actions based on current conditions.
Model-Based Reflex Agent	Uses an internal model of the environment.
Goal-Based Agent	Selects actions to achieve a goal.
Utility-Based Agent	Selects actions based on usefulness or preference.
Learning Agent	Improves its behavior through experience.

Simple Reflex Agent

Simple explanation: It reacts to the current situation.

Example: A thermostat switches on the heater when the temperature is low.

Model-Based Agent

Simple explanation: It remembers the current state and uses it to understand the situation.

Example: A robot remembers which rooms it has already visited.

Goal-Based Agent

Simple explanation: It chooses actions based on what it wants to achieve.

Example: A navigation agent finds a route to a destination.

Utility-Based Agent

Simple explanation: It chooses the action that provides the best result according to its preferences.

Example: A navigation agent selects a route that balances distance, traffic, and cost.

Learning Agent

Simple explanation: It improves its behavior using experience.

Example: A robot learns how to navigate more efficiently.

6.3 General purpose versus task-specific agents

General Purpose Agent	Task-Specific Agent
Performs multiple types of tasks.	Performs a particular task.
Can use different tools.	Usually uses a limited set of tools.
More flexible.	Specialized for its application.
Example: General AI assistant.	**Example:** Spam detection system.

Important: A General Purpose Agent may still have limitations. It can make mistakes, misunderstand instructions, or require human approval for important actions.

7. LANGUAGE MODELS AS TOOL MAKERS (LATM)

7.1 What is LATM?

Simple explanation:

LATM is nothing but allowing an AI language model to create small tools that help it solve problems.

Imagine you need to calculate the total price of 100 products.

Instead of manually calculating every product, you create a small calculator program and use it whenever required.

LATM applies a similar idea to AI.

The language model creates a reusable tool for a problem and uses that tool to perform future tasks.

Technical explanation:

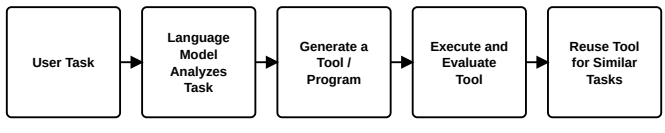
Language Models as Tool Makers (LATM) is an approach in which a language model generates executable tools or programs that can be used to solve tasks.

The model can create a tool, evaluate its usefulness, and reuse it when similar tasks occur.

This reduces repeated reasoning and can improve efficiency.

7.2 Working of LATM

DIAGRAM



Working of LATM

7.3 Example

Suppose an AI agent frequently needs to convert Celsius temperatures into Fahrenheit.

The language model creates a Python tool:

```
def celsius_to_fahrenheit(c):
    return (c * 9 / 5) + 32
```

When the agent needs to convert a temperature, it can call this tool instead of generating the calculation repeatedly.

Advantages

1. Reduces repeated reasoning.
2. Creates reusable tools.
3. Improves task efficiency.
4. Supports complex problem-solving.
5. Allows customized tools for specific tasks.

Applications

- Mathematical calculations.
- Data processing.
- Code generation.
- Automation.
- Scientific computing.
- Repeated business workflows.

Important limitation

Generated tools must be tested and validated before execution because incorrect or unsafe code can produce wrong results or unwanted actions.

8. EMERGING TRENDS IN AGENTIC AI

8.1 What are Emerging Trends?

Simple explanation:

Emerging trends are nothing but new technologies and developments that are becoming important in Agentic AI.

AI agents are moving from simple question-answering systems toward systems that can perform complex tasks using tools, memory, reasoning, and autonomous actions.

8.2 Important trends

1. Autonomous AI Agents

Simple explanation: AI systems that can complete tasks with less human involvement.

Technical explanation: Autonomous agents use perception, planning, reasoning, and action mechanisms to achieve goals independently within defined boundaries.

Example: An agent that monitors application failures and creates an issue report.

2. Multi-Agent Systems

Simple explanation: Multiple AI agents work together to complete a task.

Technical explanation: A multi-agent system consists of multiple intelligent agents that communicate, coordinate, and divide responsibilities to solve complex problems.

Example:

- Research agent collects information.
- Analysis agent studies the information.
- Writing agent prepares a report.

3. Multimodal AI Agents

Simple explanation: AI agents that understand different types of information, such as text, images, audio, and video.

Technical explanation: Multimodal agents integrate multiple input and output modalities to perceive and interact with their environment.

Example: An agent analyzes an image and answers questions about it.

4. AI Agents with Robotics

Simple explanation: AI agents are used to control physical robots.

Technical explanation: Robotic agents combine perception, planning, control, and action mechanisms to interact with physical environments.

Example: A warehouse robot moves packages.

5. Tool-Using AI

Simple explanation: AI agents use external software to perform tasks.

Technical explanation: Tool-using agents interact with APIs, databases, browsers, and other computational resources to extend their capabilities beyond language generation.

Example: An agent uses a calculator and a database to solve a business problem.

6. Long-Term Memory

Simple explanation: AI agents remember information across different tasks and conversations.

Technical explanation: Persistent memory systems store and retrieve information to support long-running tasks and personalized interactions.

Example: A personal assistant remembers previous project decisions.

7. AI Agent Frameworks

Simple explanation: Frameworks provide ready-made components for building AI agents.

Technical explanation: Agent frameworks provide abstractions for planning, tool integration, communication, memory, and execution.

Examples: AutoGen, CrewAI, and related agent development frameworks.

8.3 Benefits of emerging trends

- Increased automation.
- Improved task completion.
- Better interaction with real-world environments.
- Support for complex workflows.

- Reduced repetitive work.
- Improved information processing.

9. ETHICAL AND SAFETY CONSIDERATIONS

9.1 What is AI Ethics?

Simple explanation:

AI ethics is nothing but making sure AI systems behave responsibly and do not harm people.

For example, an AI agent should not reveal private information, perform dangerous actions, or make unfair decisions.

Technical explanation:

AI ethics refers to the principles and practices used to ensure that AI systems are fair, transparent, accountable, privacy-preserving, and beneficial to humans.

AI safety focuses on preventing unintended behavior, harmful actions, and failures during the operation of AI systems.

9.2 Major ethical and safety issues

1. Privacy

Simple explanation: AI should not expose personal information without permission.

Technical explanation: Privacy requires protecting sensitive data from unauthorized access, disclosure, or misuse.

Example: An AI agent should not share a user's private medical information with another person.

2. Bias and Fairness

Simple explanation: AI should not unfairly treat people differently.

Technical explanation: Bias occurs when an AI system produces systematically unfair outcomes due to training data, model design, or decision-making processes.

Example: A recruitment agent unfairly rejects candidates based on gender.

3. Transparency

Simple explanation: Users should understand how an AI system makes important decisions.

Technical explanation: Transparency involves providing understandable information about the system's behavior, decisions, limitations, and relevant reasoning.

Example: A loan approval system explains the main factors behind its decision.

4. Accountability

Simple explanation: Someone must be responsible when an AI system makes a mistake.

Technical explanation: Accountability establishes responsibility for AI system development, deployment, monitoring, and consequences.

Example: An organization investigates an incorrect AI-generated financial decision.

5. Hallucination

Simple explanation: AI may confidently provide information that is incorrect.

Technical explanation: Hallucination occurs when a language model generates unsupported or factually incorrect information.

Example: An AI agent invents a research paper that does not exist.

6. Security

Simple explanation: AI systems must be protected from attackers.

Technical explanation: Security involves preventing unauthorized access, malicious instructions, data theft, and unsafe tool execution.

Example: A malicious user attempts to make an agent reveal confidential information.

7. Human Oversight

Simple explanation: Humans should control important decisions made by AI.

Technical explanation: Human oversight ensures that high-impact or irreversible actions are reviewed, approved, or supervised by authorized people.

Example: An AI agent prepares a bank transfer, but a human must approve it before execution.

8. Reliability

Simple explanation: AI should work correctly and consistently.

Technical explanation: Reliability requires testing, monitoring, error handling, and safeguards against unexpected behavior.

Example: A medical support agent must not provide dangerous recommendations without appropriate validation.

9.3 AI Safety Mechanisms

Mechanism	Purpose
Human approval	Prevents unauthorized important actions.
Access control	Limits what the agent can access.
Input validation	Rejects invalid or malicious inputs.
Output validation	Checks generated information.
Monitoring	Detects errors and unsafe behavior.
Testing	Evaluates correctness and reliability.
Logging	Records agent activities for investigation.

Human-in-the-loop

Simple explanation: A human checks or approves the AI's work before an important action is completed.

Technical explanation: Human-in-the-loop is a control mechanism in which human judgment is incorporated into an AI workflow to supervise decisions, validate outputs, or authorize actions.

10. APPLICATIONS OF AUTONOMOUS AI SYSTEMS

10.1 What are Autonomous AI Systems?

Simple explanation:

An autonomous AI system is nothing but a system that can perform tasks on its own after receiving a goal.

For example, a robot vacuum cleaner can detect obstacles, choose a route, and clean a room without requiring the user to control every movement.

Technical explanation:

Autonomous AI systems are intelligent systems that perceive their environment, reason about available information, make decisions, and execute actions with limited or no continuous human intervention.

10.2 Major applications

1. Healthcare

AI agents can:

- Analyze medical information.
- Assist in appointment scheduling.
- Support medical document processing.
- Monitor patient information.
- Help healthcare professionals make decisions.

Example: A healthcare assistant retrieves patient records and prepares a summary for a doctor.

2. Smart Manufacturing

AI agents can:

- Monitor connected machines.
- Detect equipment failures.
- Analyze sensor data.
- Schedule maintenance.
- Improve production efficiency.

Example: An agent detects abnormal machine temperature and generates a maintenance alert.

3. Transportation

AI agents can:

- Analyze traffic conditions.
- Plan routes.
- Assist autonomous vehicles.
- Monitor vehicle systems.
- Improve transportation efficiency.

Example: A navigation agent selects a route based on traffic conditions.

4. Education

AI agents can:

- Explain concepts.
- Generate practice questions.
- Analyze student performance.
- Recommend learning materials.
- Provide personalized learning support.

Example: An educational agent identifies weak topics and recommends suitable exercises.

5. Software Development

AI agents can:

- Generate code.
- Review code.

- Detect errors.
- Execute tests.
- Assist in debugging.
- Prepare documentation.

Example: A coding agent identifies a bug and suggests a correction.

6. Finance

AI agents can:

- Analyze financial data.
- Detect suspicious transactions.
- Generate reports.
- Assist with customer support.
- Monitor financial activities.

Example: An agent detects an unusual transaction and alerts the security team.

7. Agriculture

AI agents can:

- Monitor soil conditions.
- Analyze crop information.
- Detect plant diseases.
- Manage irrigation.
- Support agricultural planning.

Example: An agent uses sensor data to recommend irrigation.

8. Customer Service

AI agents can:

- Answer customer questions.
- Retrieve order information.
- Track complaints.
- Create support tickets.
- Escalate difficult cases.

Example: A customer support agent retrieves an order status and explains the delivery details.

11. PYTHON-BASED MEMORY AND RAG DEMONSTRATION

This example demonstrates the basic idea of storing information and retrieving it when needed.

It is a simple educational example, not a complete production RAG implementation.

Example: Basic memory system

```
memory = {}

def remember(key, value):
    memory[key] = value

def recall(key):
    return memory.get(key, "Information not found")

remember("subject", "Agentic AI")
remember("language", "Python")

print(recall("subject"))
print(recall("language"))
```

Output

```
Agentic AI
Python
```

Explanation

- `memory = {}` creates an empty dictionary.
- `remember()` stores information.
- `recall()` retrieves information.
- `memory.get()` searches for the requested information.
- The agent can reuse previously stored information.

This demonstrates the basic concept of memory, but the information exists only while the Python program is running.

12. IMPORTANT DIFFERENCES FOR EXAM PREPARATION

Short-term Memory vs RAG vs MemGPT

Short-term Memory	RAG	MemGPT
Maintains current context.	Retrieves relevant external information.	Manages memory beyond the active context.
Used for immediate tasks.	Used for knowledge retrieval.	Used for long-running interactions.
Usually temporary.	Uses external knowledge sources.	Uses active context and external memory.
Example: Current conversation.	**Example:** Searching a PDF.	**Example:** Managing a long conversation.

General Purpose Agent vs LATM

General Purpose Agent	LATM
Designed to perform different tasks.	Focuses on generating reusable tools.
Uses reasoning, memory, planning, and tools.	Uses a language model to create tools or programs.
Example: An assistant that researches and writes reports.	**Example:** An agent generates a calculator tool.

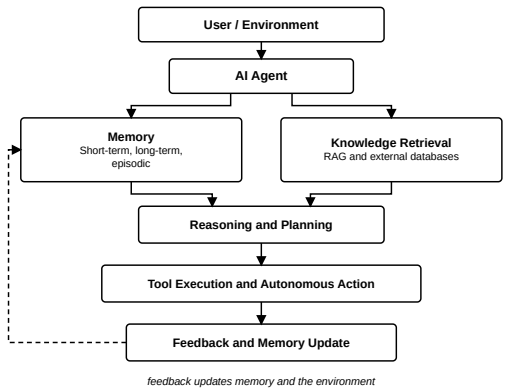
RAG vs Normal Language Model

Normal Language Model	RAG
Answers using its learned knowledge and provided context.	Retrieves external information before answering.
May not know updated information.	Can access updated external information.
May generate unsupported facts.	Retrieved context can improve factual grounding.
Does not inherently search external documents.	Includes a retrieval mechanism.

13. IMPORTANT DIAGRAM TO PRACTICE

For a 10-mark question on memory systems, RAG, or autonomous AI, practice drawing this diagram.

DIAGRAM



Important diagram to practice